



Advanced Classical Mechanics

Lecture 2: Preliminaries

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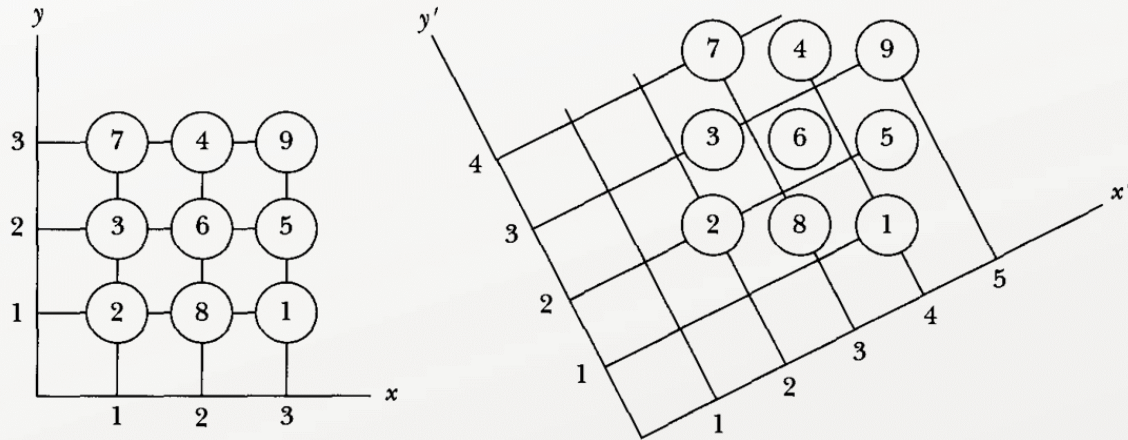
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Physical scalars

Physical quantities that are completely determined by the **specification of one numerical value and a unit** are called **physical scalars** (e.g., mass, temperature, energy, wavelength). By definition, scalars are quantities whose values are **independent of coordinate systems**.

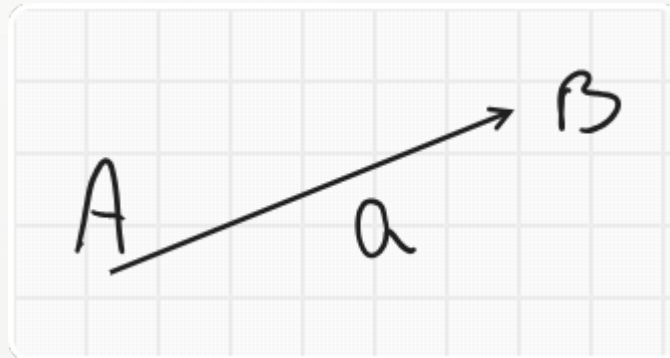


Consider, for example, a collection of 9 point masses in 2 dimensions. The masses of these point objects remain unchanged if one move from one Cartesian coordinate system on the left to a rotated Cartesian coordinate system on the right.

Physical vectors

Quantities that need the specification of a **direction** (in addition to a **numerical value** and a **physical unit**) to be completely described are called **vectors** (e.g., force, velocity, acceleration, torque).

A vector may be represented geometrically by an oriented distance, i.e., by a distance associated with a direction.



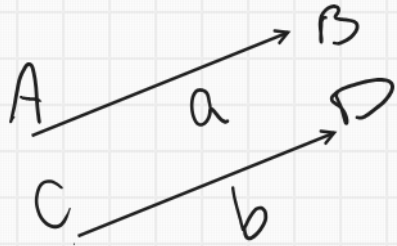
An example of a vector pointing from point A to point B.

Problem 1. We also define scalar quantities in mathematics and computer science. How do they differ from physical scalars?

Problem 2. The mass of an object is a scalar. Is the weight of an object also a physical scalar (independent of the coordinate system)? Why?

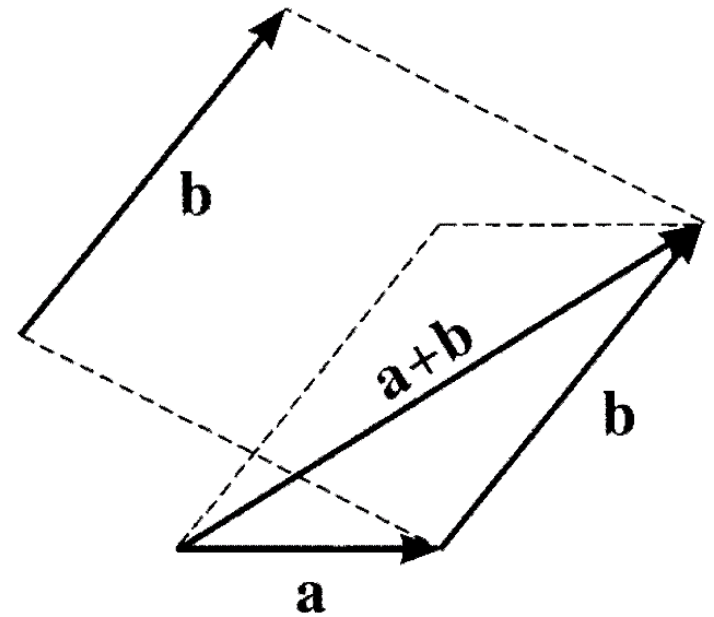
Properties of vectors

Definition. Two vectors are called **equal** if and only if the following conditions hold:



- 1) $|a| = |b|$: equal absolute values.
- ② $a \uparrow \uparrow b$: parallel directions.

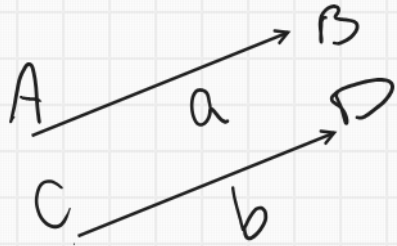
Addition of vectors. If two vectors a and b are added, the initial point of the one vector is brought by a parallel shift to coincide with the endpoint of the other one. The sum $a + b$, also called the **resultant**, corresponds to the distance from **the initial point of the first vector to the endpoint of the second one**. This sum may also be found as the diagonal of the parallelogram formed by a and b .



Addition of the vectors a and b .

Properties of vectors

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Properties of vector addition.

1) **commutation:** $a + b = b + a$

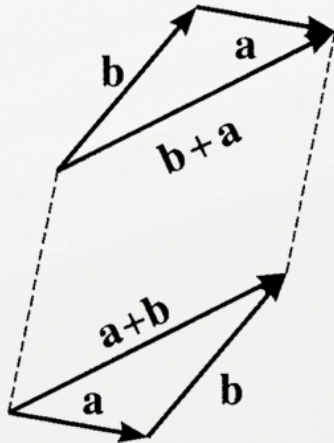


Illustration of the commutativity of the addition of vectors.

2) **association:** $(a + b) + c = a + (b + c)$

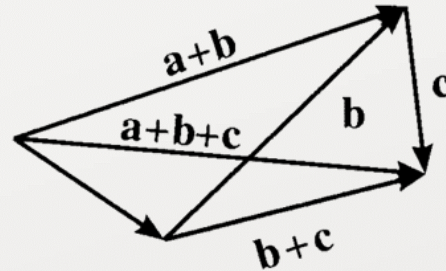
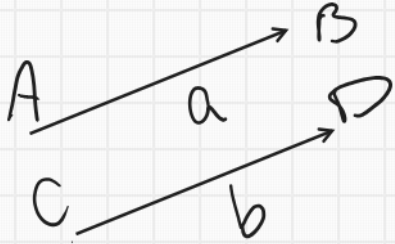


Illustration of the associativity of the addition of vectors.

Properties of vectors

Definition. Two vectors are called **equal** if and only if the following conditions hold:



- 1) $|a| = |b|$: equal absolute values.
- ② $a \uparrow \uparrow b$: parallel directions.

Vector subtraction.

The difference of two vectors $\mathbf{a}$ and $\mathbf{b}$ is simply defined in terms of addition as, $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$

Zero (Null) vector. The vector difference $\mathbf{a} - \mathbf{a}$ is denoted as **zero vector** (or **null vector**). The zero vector has **zero magnitude**. It is **orientation-less**.

Properties of vectors

Multiplication of a vector by a scalar. The product $p\mathbf{a}$ of a vector $\mathbf{a}$ by a scalar p , where p is a real number, is understood as the vector having the same orientation as $\mathbf{a}$, and the magnitude $|p\mathbf{a}| = |p| \cdot |\mathbf{a}|$.

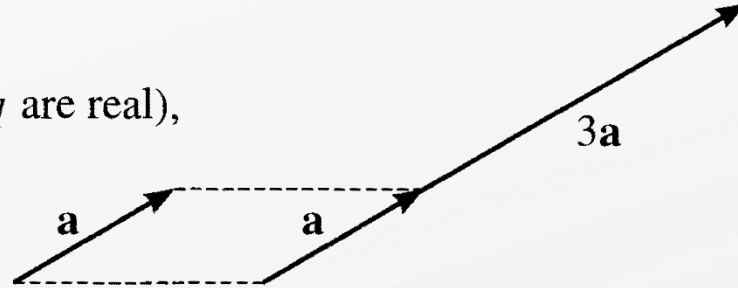
Rules of calculation:

$$q(p\mathbf{a}) = p(q\mathbf{a}) = qp\mathbf{a} \quad (\text{where } p \text{ and } q \text{ are real}),$$

$$(p + q)\mathbf{a} = p\mathbf{a} + q\mathbf{a},$$

$$p(\mathbf{a} + \mathbf{b}) = p\mathbf{a} + p\mathbf{b}.$$

These rules are immediately intelligible and don't need any further explanation.



The multiplication of a vector $\mathbf{a}$ by a scalar p (in this case, $p = 3$).

The Scalar Product (dot product)

Some physical quantities like force and path are **oriented** and are represented by vectors ($\mathbf{F}$ and $\mathbf{s}$ here). Some others, like the mechanical work W performed by a force $\mathbf{F}$ along a straight path $\mathbf{s}$ are scalars that are defined in terms of vectors, for example,

$$w = F s \cos(\phi) = |\mathbf{F}| |\mathbf{s}| \cos(\phi),$$

where ϕ is the angle enclosed by $\mathbf{F}$ and $\mathbf{s}$. W by itself, although originating from two vectors, is a scalar quantity.

Therefore, with a view on physical applications of this kind, we define the **scalar product** $\mathbf{a} \cdot \mathbf{b}$ of two vectors as,

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}| \cdot \cos(\phi),$$

where ϕ is the angle enclosed by $\mathbf{a}$ and $\mathbf{b}$. The scalar product result is a real number.

Expressed by words, the **scalar product** is defined as the **magnitude of $\mathbf{a}$ multiplied by the projection of $\mathbf{b}$ onto $\mathbf{a}$, or vice versa.**

The Scalar Product (dot product)

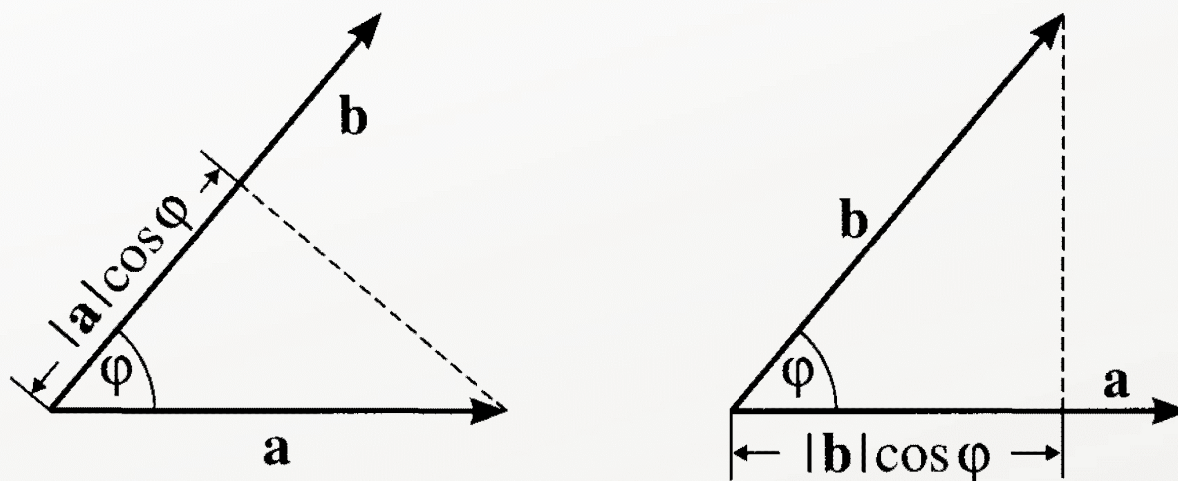


Illustration of the scalar product.

The visual meaning of the scalar product:

magnitude of the projection of $\mathbf{b}$ onto $\mathbf{a}$ multiplied by $|\mathbf{a}|$, or
magnitude of the projection of $\mathbf{a}$ onto $\mathbf{b}$ multiplied by $|\mathbf{b}|$.

The Scalar Product (dot product) properties

1. Commutativity,

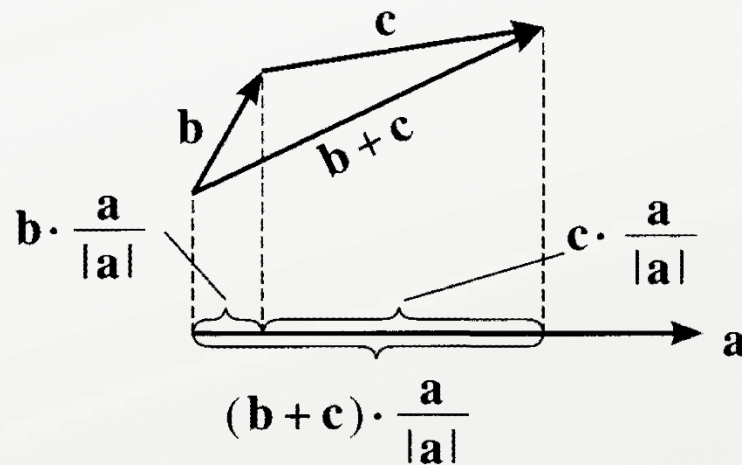
$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$

2. Associativity,

$$p \cdot (\mathbf{a} \cdot \mathbf{b}) = (p\mathbf{b}) \cdot \mathbf{a}$$

3. Distributivity,

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$$



Problem 3. When the dot product result maximum?

Problem 4. When the dot product result minimum?

Problem 5. When the dot product result zero?

Problem 6. (extra credit) Illustration of the distributivity property of dot product in three dimensions.

The Unit vectors

Unit vectors are understood as vectors of **magnitude one**.

If $\mathbf{a} \neq \mathbf{0}$, then,

$$\mathbf{e} = \frac{\mathbf{a}}{|\mathbf{a}|},$$

is a unit vector pointing along the vector $\mathbf{a}$.

Problem 7. Based on the above that the magnitude of a unit vector is one.

A possibility frequently used in physics is to assign a direction to a scalarly formulated equation by the unit vector. For example, the gravitational force has the magnitude,

$$F = G \frac{mM}{r^2},$$

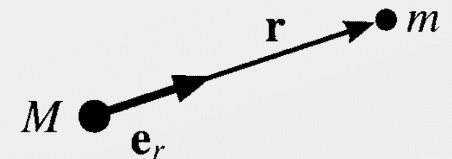
The force is acting along the connecting line between the two masses M and m , hence,

$$\mathbf{F} = G \frac{mM}{r^2} \frac{\mathbf{r}}{|\mathbf{r}|},$$

$\mathbf{F}$ is the force applied by the mass M to the mass m .

Its direction is given by the extra ratio at the end of equation.

Hence it is acting toward the mass M .



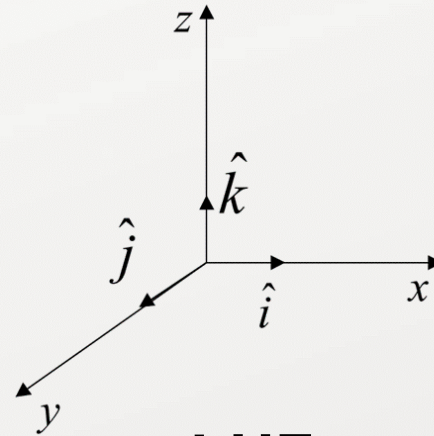
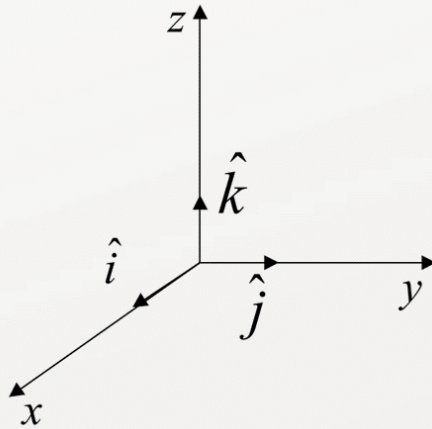
The unit vector pointing from the big mass to the small mass is $\mathbf{e}_r = \mathbf{r}/|\mathbf{r}|$.

The Cartesian Unit vectors

The unit vectors pointing along the positive x-, y-, and z-axes of a Cartesian coordinate frame are defined as follows:

1. $\mathbf{e}_1$ (in x-direction) or also $\mathbf{i}$;
2. $\mathbf{e}_2$ (in y-direction) or also $\mathbf{j}$;
3. $\mathbf{e}_3$ (in z-direction) or also $\mathbf{k}$.

There exist two kinds of Cartesian coordinate frames, namely **right-handed frames** and **left-handed frames**.



We shall always use only right-handed frames (**RHF**) in our lectures.

The Cartesian Unit vectors: Orthonormality relations

Consider the properties of the Cartesian unit vectors concerning the formation of scalar products. Since the enclosed angle is each a right one, the following relations hold:

$$\begin{aligned} \mathbf{i} \cdot \mathbf{i} &= \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1, \\ \mathbf{i} \cdot \mathbf{j} &= \mathbf{j} \cdot \mathbf{k} = \mathbf{i} \cdot \mathbf{k} = 0, \end{aligned}$$

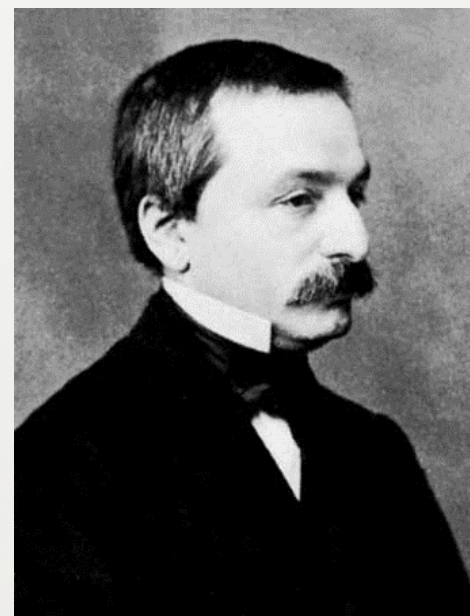
These relations are combined by defining,

$$\mathbf{e}_\mu \cdot \mathbf{e}_\nu = \delta_{\mu\nu} = \begin{cases} 0 & \text{if } \mu \neq \nu \\ 1 & \text{if } \mu = \nu \end{cases}$$

Where $\delta_{\mu\nu}$ called the **Kronecker symbol**.

For the three-dimensional space, μ , and ν run from 1 to 3, corresponding to $\mathbf{e}_1 = \mathbf{i}$, $\mathbf{e}_2 = \mathbf{j}$, $\mathbf{e}_3 = \mathbf{k}$.

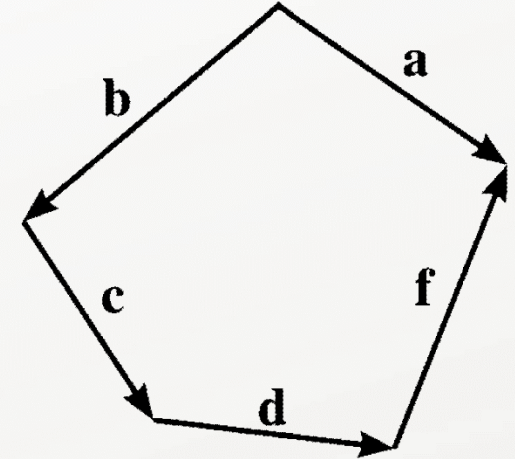
Leopold Kronecker, b. Dec. 7, 1823, Liegnitz (Legnica)-d. Dec. 29, 1891, Berlin. Kronecker was a rich private person who moved to Berlin in 1855. He taught for many years at the university there, without having a chair. Only in 1883, after retirement of his teacher and friend Kummer, he took a professorship. His most important publications concern arithmetics, theory of ideals, number theory, and elliptic functions. Kronecker was the leading representative of the Berlin School of thought, which claimed the necessity of arithmetization of the entire mathematics.



Component Representation of a Vector

The vector $\mathbf{a}$, which is uniquely represented by the sum of the vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{f}$ is called the **linear combination** of the **vectors** (e.g., $\mathbf{b}, \mathbf{c}, \mathbf{d}$, and $\mathbf{f}$).

The terms "**vectors**" and their "**linear combination**" thus graphically form a closed polygon, the **vector polygon**. One may, of course, build a the linear combination from given vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ that yields the arbitrary (but fixed) vector $\mathbf{a}$.



According to the definition introduced above, the vector $\mathbf{a}$ then must be a linear combination of the vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$; thus,

$$\mathbf{a} = q_1 \mathbf{b} + q_2 \mathbf{c} + q_3 \mathbf{d}$$

The vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ must be linearly independent, that is, none of the three vectors may be represented by the other two vectors. Otherwise not every arbitrary vector $\mathbf{a}$ could be represented as a combination of the three basic vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ (in 3 dimensions). If, for example, $\mathbf{d}$ could be expressed by $\mathbf{b}$ and $\mathbf{c}$, hence $\mathbf{d} = \alpha \mathbf{b} + \beta \mathbf{c}$, then $\mathbf{a} = (q_1 + q_3 \alpha) \mathbf{b} + (q_2 + q_3 \beta) \mathbf{c}$ would always be confined to lie in the plane spanned by $\mathbf{b}$ and $\mathbf{c}$. But an arbitrary vector $\mathbf{a}$ generally does not lie in this plane (e.g., points out of this plane).

One says: The base $\mathbf{b}, \mathbf{c}$ is incomplete for arbitrary vectors $\mathbf{a}$.

In the three-dimensional space one therefore always needs **three basic vectors** that are **linearly independent** (i.e., cannot be expressed by each other).

Component representation of a vector in Cartesian coordinates

Any vector of the three-dimensional space may be represented as a linear combination of the Cartesian unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$. This representation leads to simple and transparent calculations, due to the **orthogonality relations**. One then has,

$$\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k} ,$$

where $a_x = \mathbf{a} \cdot \mathbf{i}, a_y = \mathbf{a} \cdot \mathbf{j}, a_z = \mathbf{a} \cdot \mathbf{k}$, are the projections of $\mathbf{a}$ onto the axes of the frame. The unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ (or $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$) are also called **base (or basis) vectors**.

Besides the representation as a sum of vectors along the unit vectors, the vector $\mathbf{a}$ still may be represented as,

$$\mathbf{a}^T = (a_x, a_y, a_z) ,$$

which is in the row (**transposed**) notation, or

$$\mathbf{a} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} ,$$

which is in the column notation.

If the base vectors are known, it is sufficient to know the three components.

Calculation of the magnitude of a vector from the components. According to the **Pythagorean theorem**, which we studied last time, the magnitude of a vector $\mathbf{a}$ is calculated from its **Cartesian components** as follows:

$$|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Component representation of a vector in Cartesian coordinates

Addition of vectors expressed by components. It follows in 3-dimensions that,

$$\begin{aligned}\mathbf{a} + \mathbf{b} &= \sum_{i=1}^3 a_i \mathbf{e}_i + \sum_{i=1}^3 b_i \mathbf{e}_i = \sum_{i=1}^3 (a_i + b_i) \mathbf{e}_i \\ &= (a_1 + b_1) \mathbf{e}_1 + (a_2 + b_2) \mathbf{e}_2 + (a_3 + b_3) \mathbf{e}_3 \\ &= (a_1 + b_1, a_2 + b_2, a_3 + b_3) .\end{aligned}$$

Here both **commutativity** as well as **associativity** of **vector addition** have been used repeatedly. Thus, the components of the sum vector are the sums of the corresponding components of the individual vectors.

Problem 8. Derive the component-wise expression for **scalar product** in three dimensions for $\mathbf{a}$ and $\mathbf{b}$ vectors as defined above and show that,

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^3 a_i b_i ,$$

Hence, **the scalar product of two vectors may be evaluated simply by multiplying the corresponding components of the vectors by each other and summing over the three products.**

Problem 9. Prove the distributivity property of the scalar product using the component-wise representation of the vectors involved.

Component representation of a vector in Cartesian coordinates

Problem 10. Addition and subtraction of vectors. An airplane appears to fly northwest at 930 km/h relative to the ground while a strong breeze is blowing from the west (toward the east) at 120 km/h relative to the ground. What are the velocity and direction of the plane's flight, assuming there is no wind deflection?

The final answer is, $v_0 = 1018.39 \frac{km}{h}$, $\phi = 40.2$ (toward west)

The position vector. A point P in space may be uniquely fixed by specifying the vector beginning at the origin of the coordinate frame and pointing to the point P as the end point. The components of this vector, the position vector, correspond to the coordinates (x, y, z) of the point P . Thus, for the position vector, which is mostly abbreviated by $\mathbf{r}$, there holds,

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} ,$$

$$|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2} ,$$

Problem 11. The angle between two vectors. Recall the definition of **scalar product**. Write an equation for the angle between two vectors $\mathbf{a}$ and $\mathbf{b}$ in terms of their Cartesian components in three dimensions.

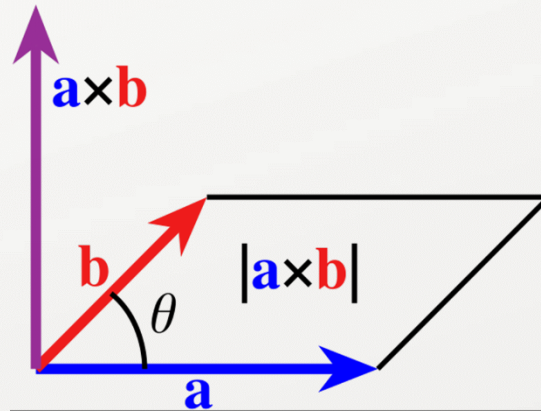
The Cross (Vector) Product (Axial Vector)

Not all physical operations can be expressed as scalar products. One may define a further product between vectors, one of which is defined as follows.

Definition. The cross product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is the vector,

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \sin(\theta) \mathbf{n}$$

where $\mathbf{n}$ is the unit vector being perpendicular to the plane fixed by $\mathbf{a}$ and $\mathbf{b}$, and pointing out of the plane as a right-handed helix when rotating the first vector of the product into the second vector. **Note that the rotation has to be performed along the shortest path.**



The geometrical interpretation of the Cross Product.

Problem 12. Under what circumstances the result of cross product is zero?

Problem 13. Express the Cartesian basis vectors ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) in terms cross products of each other.

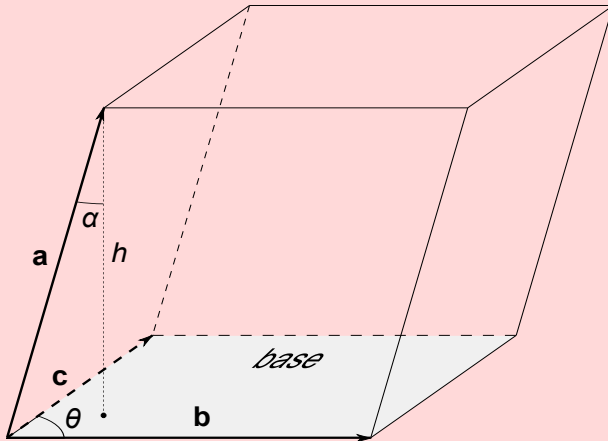
The Cross (Vector) Product (Axial Vector) properties

The cross-product has the following properties,

1. anti-commutativity: $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
2. distributivity: $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
3. no associativity: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$
4. scalar multiplication: $p\mathbf{a} \times \mathbf{b} = \mathbf{a} \times p\mathbf{b} = p(\mathbf{a} \times \mathbf{b})$

Problem 14. Prove the distributivity property of the Cross Product. (Hint: Write the vectors in terms of Cartesian basis vectors $(\mathbf{i}, \mathbf{j}, \mathbf{k})$).

Problem 15. Consider the following parallelepiped, formed by the three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Express the volume of the parallelepiped as product(s) of these vectors.



Problem 16.

- a. The vector $(1, a, b)$ is perpendicular to the two vectors $(4, 3, 0)$ and $(5, 1, 7)$. Find a and b .
- b. Evaluate in Cartesian coordinates the vector product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = (1, 7, 0)$ and $\mathbf{b} = (1, 1, 1)$.
- c. Show that $(\mathbf{a} \times \mathbf{b})^2 = a^2 b^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

Problem 17. Prove the **double cross-product** identity: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$

Problem 18. Prove the alternate **double cross-product** identity: $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{c})$

